7 Molybdenum–99 ^{99}Mo is a key radioisotope used to produce Technetium–99m ^{99m}Tc, one of the most widely used tracer in medical imaging like bone and heart scans. In a nuclear reactor, Molybdenum–99 is produced as a fission product when Uranium–235 ^{235}U absorbs a slow neutron.

(a) A typical fission reaction producing Molybdenum-99 is given by the equation:

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{99}_{42}Mo + {}^{134}_{z}Sn + x\,{}^{1}_{0}n$$

(i) Determine the values of the proton number z and the integer x representing the number of emitted neutrons.

z = ...

x = ...

[2]

(ii) The binding energies per nucleon for the relevant nuclides are given below:

^{235}U : 7.59 MeV
^{99}Mo : 8.61 MeV
^{134}Sn : 8.45 MeV

Calculate the energy released in this fission reaction.

energy released = ... MeV [2]

(iii) This nuclear fission reaction is also called a *chain* reaction.

Suggest what is meant by a *chain* reaction.

.....

..... [1]

Examiner's Use

(b) Once Molybdenum–99 is produced, it undergoes β^- decay to produce Technetium–99m ^{99m}Tc. Because ^{99m}Tc has a short half-life, it cannot be stored long-term. Instead, it is produced in hospitals using a portable generator containing Molybdenum–99.

The following data is provided:

Mass of Molybdenum–99 ^{99}Mo	: 98.90771 *u*
Mass of Technetium-99m ^{99m}Tc	: 98.90641 *u*
Mass of an electron	: 0.00055 *u*

(i) Radioactive decay is a *random* and *spontaneous* process.

Explain what the following terms mean.

1. *random*

.....

..... [1]

2. *spontaneous*

.....

..... [1]

(ii) Write down a balanced nuclear equation representing the β^- decay of Molybdenum–99 to Technetium–99m, including any fundamental particles emitted.

[1]

(iii) Determine the energy released in MeV during the decay of Molybdenum–99.

energy released = ... MeV [3]

(c) The half-life of Molybdenum–99 and Technetium–99m is 66 hours and 6.0 hours respectively.

(i) Define *decay constant*.

.....

..... [1]

(ii) A fresh medical generator arrives at a hospital containing a sample of pure Molybdenum–99 with an initial activity of 12.0 GBq.

Determine the initial mass of Molybdenum–99 nuclei present.

initial mass = ... kg [3]

(iii) Hence determine the mass of Molybdenum–99 nuclei present after 30 days.

mass = ... kg [2]

(d) Technetium–99m is attached to a carrier molecule that targets a specific part of the body when injected into a patient. When Technetium–99m transitions to its ground state Technetium–99 ^{99}Tc, it emits a γ–ray photon. Technetium–99 has a much longer half-life of 2.11×10^5 years.

(i) Explain why Technetium–99m is highly suitable as a medical tracer.

.....

..... [1]

(ii) Explain why Technetium–99 poses negligible radiation risk to the patient post-scan.

.....

..... [1]

(iii) In contrast to medical tracers, industrial tracers that track fluid flow or leaks in deep underground chemical pipelines utilise radioisotopes with longer half-lives.

Suggest why a longer half-life is necessary for industrial pipeline applications.

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..... [1]

[Total: 20]